Reg. No. | | | | | | | | | | | [17CIC43]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

OPERATING SYSTEMS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions. Choosing at least **TWO** questions from each part.
2. Any missing data may be suitably assumed.

PART-A

1. **a.** Briefly explain the objectives of Operating system. Also list the services of operating system. **05**

b. What is thread? List the key benefits of Threads. **03**

c. With a neat diagram, explain the working of a Virtual machine. What are the advantages and disadvantages of a Virtual machine? **06**

2. **a.** With a neat diagram, explain five state model of a Process. **05**

b. Consider the following data with burst time given in milliseconds. **09**

Process	Burst time	Priority
P1	10	3
P2	1	1
P3	2	3
P4	1	4
P5	5	2

The processes are arrived in the order P1, P2, P3, P4, and P5 all at time 0.

i. Draw Gantt charts for the execution of these processes using FCFS, SJF, Non pre-emptive priority and RR (quantum=1) scheduling.

ii. What is the Turnaround time and Waiting time of each process for each of the scheduling algorithm?

3. **a.** What are Semaphores? Explain Binary and Counting Semaphores with an example. **05**

b. Write Peterson's algorithm for 2 Process Synchronization to Critical Section problem and explain briefly. **04**

c. Give the Monitor solution to the Dining Philosopher problem. **05**

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4. a. Consider the following snapshot of a system. **06**

Processes	Allocation			MAX			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			

Answer the following questions using Banker's algorithm.

i. What is the content of the Need matrix?

ii. Is the system in a safe state?

iii. If a request from process P0 arrives for (0, 2, 0) can the request be granted immediately?

b. With a neat sketch, explain Resource allocation graph showing the situation of Deadlock and no Deadlock, for multiple instances of the resources. **04**

c. Write Deadlock Detection algorithm. **04**

PART-B

5. a. Explain different Memory management requirements. **04**

b. Differentiate **06**

i. Internal and External fragmentation

ii. Page and Frame

iii. First fit and Best fit algorithm

c. Explain Paging with neat diagram. **04**

6. a. Consider the following Page reference string: **09**
1,2,3,4,2,1,5,6,2,1,3,7,6,3,2,1,2,3,6. Find out the number of page faults if there are 4 page frames using the following page replacement algorithm.

i. LRU

ii. FIFO

iii. Optimal

b. What is Thrashing? What are the causes of Thrashing? **05**

7. a. Exemplify the different types of Directory structures, with examples and mention their advantages and disadvantages. **07**

b. With supporting diagrams, explain Linked and Index method of allocating disk space. **07**

8. a. Discuss the different components of Linux system with a neat sketch. **05**

b. Explain Linux Virtual file system. **04**

c. Explain the memory management of Linux Operating system. **05**